

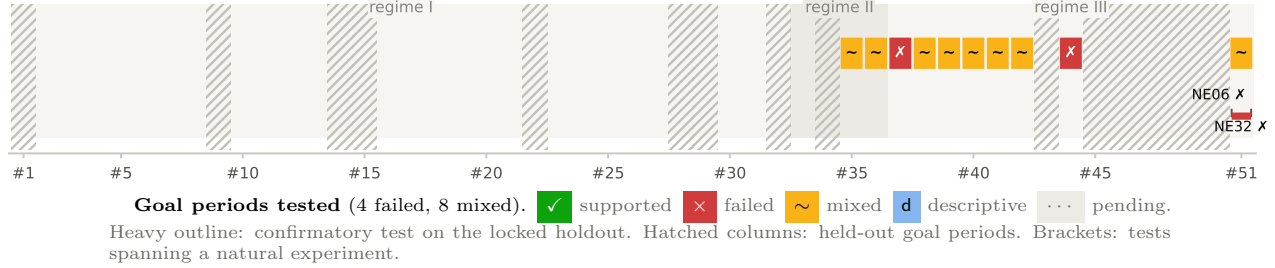
H13: Model families carry their own fields, and couple by family

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Model families carry a content field that is mostly style, and content couples more within labs at the read-out call; talk does not.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



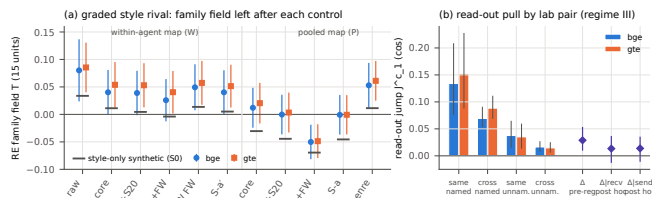
WHAT HAPPENED

The data is 15 counted non-reserved units (#35–#44, #51; $N = 10\text{--}27$). The state is each agent’s daily content vector with the day’s common field removed. Same-lab agents share a content direction in 9/15 units (RE $T = 0.08$ [0.02, 0.14]), stable across periods. **Round 2** grades the style rival. A within-agent map of 20 formatting features and 50 function words keeps **32% (bge) to 47% (gte)** of the field: $T = 0.026$ [−0.01, 0.06] and 0.040 [0.002, 0.08]. Round 1’s pooled control was biased downward. Newcomers start at their lab’s field ($a(1) = 0.21$, $p = 0.005$) and drift away. At the read-out call, a message pulls a **same-lab reader’s content 1.7× more** (0.067 vs 0.039; 8/8 units). About half of that is lab-level susceptibility (post hoc). Talk shows no family term.

PHYSICAL MODEL

Model 11 (vector spins). $v_{i,d}$ is agent i ’s mean whitened chat embedding on day d , and m_d is the day mean. h_f is the field of lab $f(i)$, a_i an agent offset, and H_i the agent field. Style rival: each statement vector z is regressed on its style features s (formatting, function words, genre held out) with a map B fitted within agents (W) or pooled (P). Coupling: J_1 is the matched-age jump in $\cos(z_B, z_m)$ between the recipient’s read-out call ($h = 1$) and the call in flight ($h = 0$), for same-lab and cross-lab pairs.

$$v_{i,d} = m_d + h_{f(i)} + a_i + \varepsilon_{i,d}, \quad T = \langle \cos H_i H_j \rangle_{\text{same}} - \langle \cos H_i H_j \rangle_{\text{cross}}$$
$$z \mapsto z - (s - \bar{s}) \hat{B}_W, \quad \rho_{\text{ns}} = T_{W3} / T_{\text{raw}}$$
$$J_1 = \sum_b w_b [\bar{y}(1, b) - \bar{y}(0, b)], \quad \Delta = J_1^{\text{same}} - J_1^{\text{cross}}$$



Observables. (a) Random-effects family field after each style control, bge (blue) and gte (orange), with 95% CIs. Dashes: the same control on a synthetic world whose family field is style only. Within-agent maps keep a third to a half of the field. Pooled maps fall below zero even in the style-only world. (b) Read-out content pull, regime III. Same-lab pairs pull more in both naming strata. Violet: the named-stratified contrast (pre-registered) and, post hoc, the contrast with the receiver’s or the sender’s lab held fixed.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Vendors differ mostly in prose, and partly in what they say. A pooled style correction removes the prose and then pushes the vendor field below zero. A within-agent correction leaves about a third of it. Vendors also differ in how strongly their models are moved by what they read. A same-vendor message moves a reader 1.7× as much as a cross-vendor one, at the next call. Half of that is because some vendors’ models are more easily moved, not in-group preference. Who replies does not depend on vendor. New models arrive sounding like their vendor and lose part of it within a week. For an operator: expect a read message to move a same-vendor model up to 1.7× as much as another vendor’s model.

H13-family-fields generated 2026-10-07 17:50 by in-fra/summaries/build_summaries.py card hypotheses/H13-family-fields/README.md ratings: H13 round-2 agent (suggested; coordinator writes v2)

RESULTS VS PREDICTIONS

Prediction	Result	Verdict
P1 family field (r1, 1b)	9/15; RE 0.080; gte 8/15, 0.085	mixed
P5, P6 no co- talk RE -0.003 ; content 1/15 movement homophily (r1, 1b)		✓
A-P1 monotone style W: $0.080 \rightarrow 0.040 \rightarrow 0.039 \rightarrow 0.026$ ladder		✓
A-P2 mostly style bge 0.32, gte 0.47; 3/15 units ($\rho_{ns} \in (0.1, 0.5]$)		✓
A-P4 not speech-act families differ in genre mix 6/15 mix		×
B-P1 newcomers start at lab $a(1) = 0.205$, $p = 0.0045$ (20 joiners)		✓
B-P2 lab alignment slope $-0.034/\text{day}$ $[-0.065, -0.003]$ does not fade		×
B-P4 newcomers con- start nearer the centroid; no trend verge on room		×
C-P1 no content fam- $\Delta = 0.029$ $[0.010, 0.054]$; 8/8 units ily coupling		×
C-P2 no talk family $\Delta\beta = -0.0002$ $[-0.0016, 0.0014]$ coupling		✓

GOAL PERIODS IN DETAIL

G35 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G36 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G37 $\times$ 1b: failed (round 1: failed; same in both embedding m...
G38 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G39 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G40 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G41 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G42 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...
G44 $\times$ 1b: failed (round 1: failed; same in both embedding m...
G51 $\sim$ 1b: mixed (round 1: mixed; same in both embedding mod...

+2 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	2	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

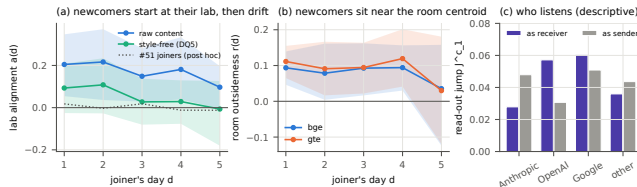
The read-out contrast mixes homophily with lab main effects; with the receiver's lab fixed it is 0.013 $[-0.013, 0.037]$. Its power at half the pull is 0.55. The non-style family field is resolved only in gte; power at a third of the field is 0.30. The newcomer drift comes from joiners before #51 (post hoc). The leave-family-out day mean was withdrawn as biased. The confirmatory run (C1–C6) has not run.

ROUND 2

Fit the read-out pull as receiver susceptibility times sender potency plus a homophily term, pooled across units with shrinkage. Add the reserved #51 tail only after Vivian's sign-off.

- **Essence**. Do vendors differ in decisions, not prose?
- **H13-R1** Policy fingerprint: $P(\text{next behavior state} \mid \text{current state, context features})$ differs by family beyond agent identity.
- **H13-R2** Families differ in susceptibility: who listens to whom (HH98).
- **H13-R3** Mixed-vendor periods out-produce single-vendor ones (descriptive, confounded; work ledger).

H13-family-fields generated 2026-10-07 17:50 by infra/summaries/build-summaries.py card hypotheses/H13-family-fields/README.md ratings: H13 round-2 agent (suggested; coordinator writes v2) (page 2)



Second figure. (a) Newcomers' lab alignment by day since joining (20 joiners; bands: joiner bootstrap). Raw content starts toward the newcomer's own lab and falls. Style-free content starts weaker. #51 joiners (dotted, post hoc) start neutral. (b) Room outsiders: positive means nearer the room centroid than incumbents. (c) Read-out pull by the receiver's and the sender's lab (point estimates): Anthropic models are the least moved receivers.